

**Important note:** \*denotes questions that may require knowledge from later chapters in order to solve them.

## I. MCQs

### 1. [AJC/Prelims 2014/P1/24]

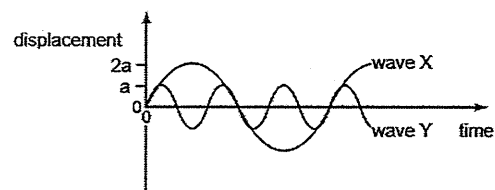
A progressive wave in a stretched string has a speed of  $2.0 \text{ m s}^{-1}$  and a frequency of 100 Hz. What is the phase difference between two points 25 mm apart?

- A zero      B  $\frac{\pi}{4}$       C  $\frac{\pi}{2}$       D  $\pi$

### 2. [AJC/Prelims 2014/P1/18]

The intensity of a progressive wave, besides being dependent on the amplitude of the wave, is also proportional to the square of the frequency.

The diagram shows two waves X and Y.

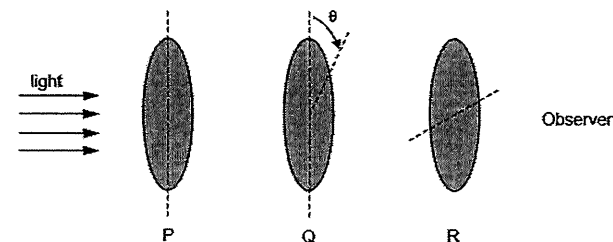


The intensity of wave X is  $I_0$ . What is the intensity of wave Y?

- A  $0.028 I_0$       B  $0.11 I_0$       C  $0.44 I_0$       D  $2.25 I_0$

### 3. [AJC/Prelims 2014/P1/20]

Polaroids P, Q and R are aligned such that the axes of polarisation of P and Q are aligned to each other and the axis of polarisation R is perpendicular to that of P and Q as shown.

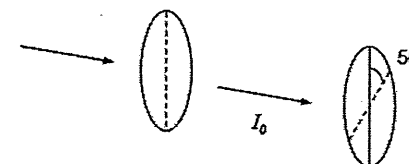


Polaroid Q is then rotated about its centre through an angle  $\theta$  until its axis of polarisation is aligned with that of R. Which of the following describes the change in light intensity seen by the observer?

- A Light intensity remains zero all the time.  
B Light intensity was maximum initially and decreases to zero.  
C Light intensity decreases to zero and then increases back to maximum.  
D Light intensity increases to maximum and then decreases back to zero.

### 4. [DHS/Prelims 2014/P1/19]

Unpolarized light is incident on a polarizer as shown below. The intensity of the light emerging from the first polarizer is  $I_0$ . The first polarizer is vertically polarized while the polarizing axis of the second polarizer is  $50^\circ$  from the vertical.

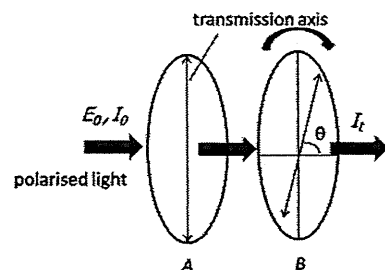


What is the intensity of light emerging from the second polarizer?

- A  $0.413 I_0$       B  $0.642 I_0$       C  $0.826 I_0$       D  $I_0$

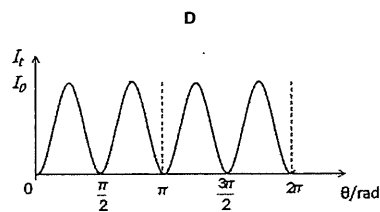
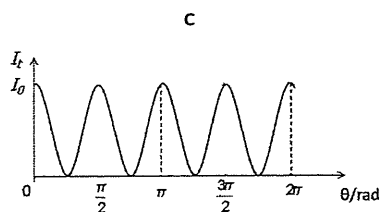
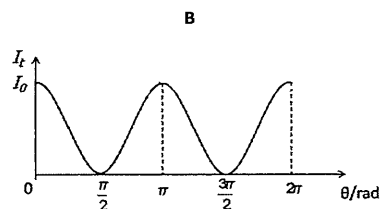
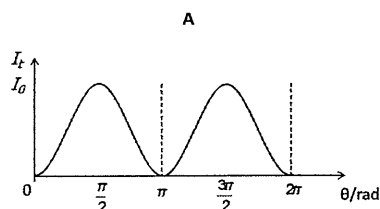
## 5. [CJC/Prelims 2014/P1/20]

The figure shows two ideal polarisers  $A$  and  $B$  where their transmission axes are initially parallel to each other.



Polarised light of amplitude  $E_0$  and intensity  $I_0$  is incident on  $A$  with its electric field vector parallel to the transmission axis. Polariser  $B$  is then rotated so that its transmission axis makes an angle  $\theta$ , as shown in the figure above.

Which of the following graphs shows how the intensity of the transmitted light  $I_t$  varies with the angle  $\theta$ ?



## 6. [DHS/Prelims 2014/P1/20]

A progressive wave is one which

- A has vibrations that are perpendicular to the direction of wave travel.
- B has vibrations that are parallel to the direction of wave travel.
- C transfers energy in the direction of wave travel.
- D transfers energy and particles in the direction of wave travel.

## 7. [IJC/Prelims 2014/P1/19]

$X$  and  $Y$  are two points on the surface of water in a ripple tank. A source of waves of constant frequency begins to generate waves which then travel past  $X$  and  $Y$ , causing them to oscillate.

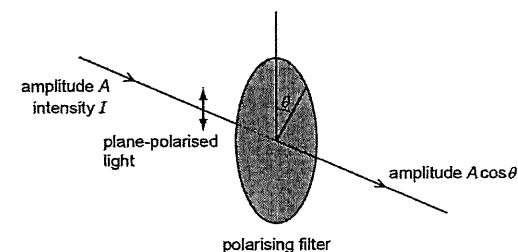


What is the phase difference between  $X$  and  $Y$ ?

- A  $45^\circ$
- B  $135^\circ$
- C  $180^\circ$
- D  $270^\circ$

## 8. [IJC/Prelims 2014/P1/20]

When plane-polarised light of amplitude  $A$  is passed through a polarising filter as shown, the amplitude of the light emerging is  $A \cos \theta$ .



The intensity of the incident beam is  $I$ .

What is the intensity of the emerging light when  $\theta$  is  $60^\circ$ ?

- A  $0.25 I$
- B  $0.50 I$
- C  $0.75 I$
- D  $0.87 I$

## 9. [IJC/Prelims 2014/P1/24]

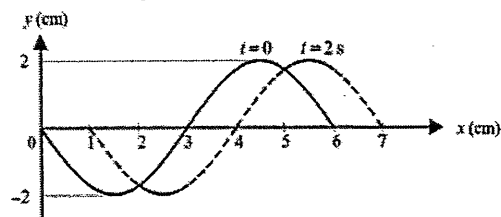
In microwave ovens, water molecules in food are set into resonance when microwaves of fixed frequency are incident upon them. This causes the molecules to receive energy and hence warm up the food.

In order to warm up the food faster, one can

- A increase the frequency of the incident microwave while keeping its amplitude fixed.
- B increase the frequency and amplitude of the incident microwave.
- C increase the amplitude of the incident microwave while keeping its frequency fixed.
- D keep both frequency and amplitude of the incident microwave the same as before but increase the frequency of the water molecules.

## 10. [JJC/Prelims 2014/P1/25]

The figure shows the positions of a travelling wave at time intervals  $t = 0$  and  $t = 2$  s.



What is the speed of the wave?

- A  $0.50 \text{ cm s}^{-1}$       B  $2.0 \text{ cm s}^{-1}$       C  $0.50 \text{ m s}^{-1}$       D  $2.0 \text{ m s}^{-1}$

## 11. [MJC/Prelims 2014/P1/17]

Alice is standing 15 m away from a speaker listening to a music broadcast. Due to a technical fault, the power of the speaker is suddenly reduced by 25%.

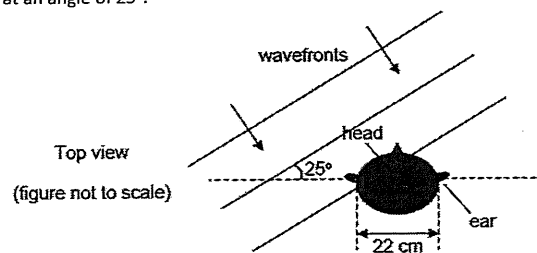
How far away from the speaker should Alice stand now in order for the music to be as loud as before? Assume that the speaker is a point source of sound.

- A 13 m      B 11 m      C 7.5 m      D 3.8 m

## 12. [MJC/Prelims 2014/P1/18]

Humans are able to detect the general direction of a sound source because sound from the source reaches the ear at slightly different times.

The figure below shows a human head with the ears 22 cm apart. Sound waves of wavelength 1.7 m from a distant source reach the head at an angle of  $25^\circ$ .

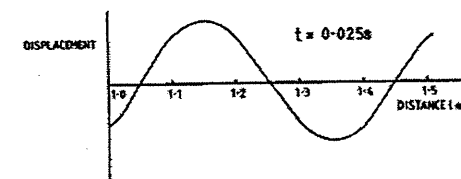
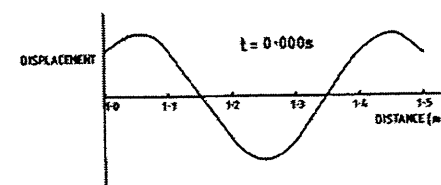


What is the phase difference between the waves reaching the left and right ear?

- A 0.34 rad      B 0.58 rad      C 0.74 rad      D 0.81 rad

## 13. [NJC/Prelims 2014/P1/16]

The diagrams show a wave in the same section of a string at two different times indicated.



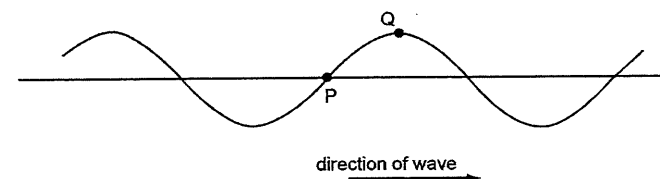
What is the longest period of the wave consistent with the above information?

- A 0.025 s      B 0.050 s      C 0.10 s      D 0.40 s

## 14. [NYJC/Prelims 2014/P1/19]

The diagram shows a transverse wave on a rope. The wave is travelling from left to right.

At the instant shown, the points P and Q on the rope have zero displacement and maximum displacement respectively.



Which of the following describes the direction of motion, if any, of the points P and Q at this instant?

|   | point P      | point Q      |
|---|--------------|--------------|
| A | downwards    | downwards    |
| B | downwards    | stationary   |
| C | upwards      | downwards    |
| D | to the right | to the right |

## 15. [RVHS/Prelims 2014/P1/17]

The wave disturbance created in a swimming pool passes vertically along the surface of the water. Two rubber ducks are placed at points X and Y along the propagation of the wave with a phase difference of  $\frac{3}{2}\pi$  radians and the waves are travelling from point X to Y.

Which row correctly describes the rubber duck at X at an instant when the rubber duck at Y is displaced downwards and moving downwards?

|   | displacement of rubber duck at X | movement of rubber duck at X |
|---|----------------------------------|------------------------------|
| A | downwards                        | downwards                    |
| B | downwards                        | upwards                      |
| C | upwards                          | upwards                      |
| D | upwards                          | downwards                    |

## 16. [RVHS/Prelims 2014/P1/18]

Ripples on the surface of water spread out in circles from a ball type dipper in a ripple tank. For one such ripple, the amplitude of the ripple at a distance 80 mm from the dipper is 2.0 mm. After a while, the power of the dipper is doubled.

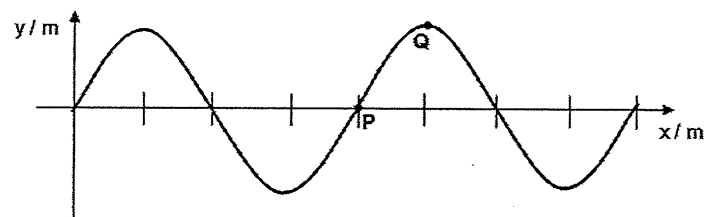
Assume that the energy of the wave is spread over the entire circumference of the ripple and no energy is lost in the propagation of the ripple.

How far would the new distance be from the dipper for the same amplitude of the ripple to be observed?

- A 40.0 mm      B 56.6 mm      C 113 mm      D 160 mm

## 17. [SAJC/Prelims 2014/P1/19]

The diagram shows a transverse wave at a particular instant. The wave is travelling to the right at a frequency of 12.5 Hz.

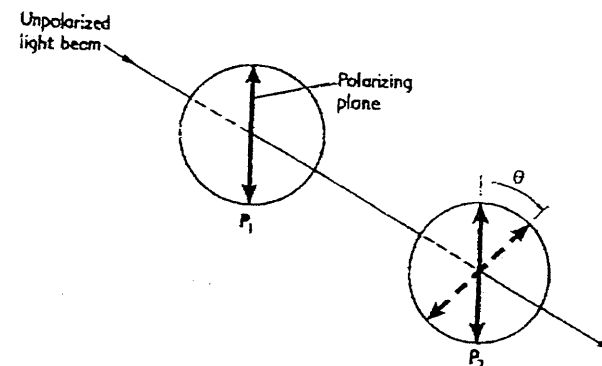


At the instant shown, the displacement at point P is zero. What is the possible time interval that would elapse before the displacement is zero at point Q?

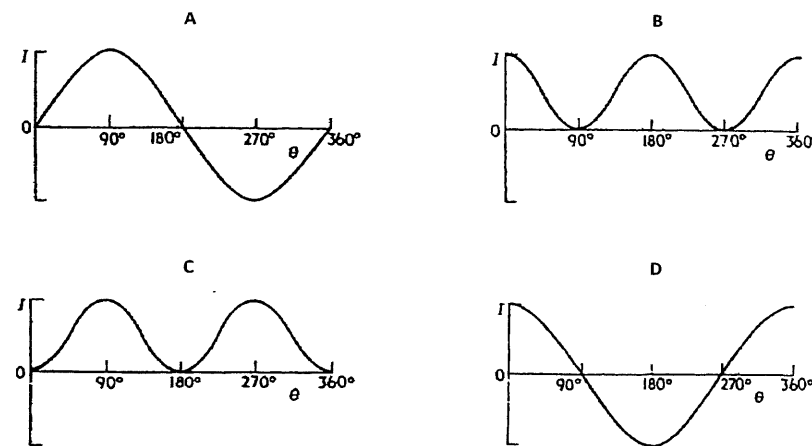
- A 0.01 s      B 0.03 s      C 0.04 s      D 0.06 s

## 18. [SAJC/Prelims 2014/P1/20]

A beam of unpolarised light passes in turn through polaroids  $P_1$  and  $P_2$  as illustrated in the figure below. The polarising plane of the polaroids are indicated by arrows. Polaroid  $P_1$  is fixed while Polaroid  $P_2$  is rotated through  $360^\circ$  from the position where  $\theta = 0^\circ$ .



Which of the following graphs best shows how the intensity  $I$  of the emergent beam varies with the angle of rotation  $\theta$ ?



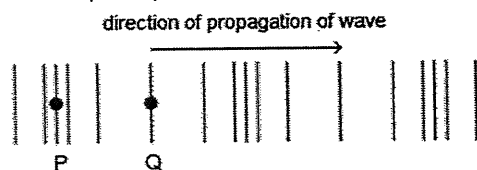
## 19. [SRJC/Prelims 2014/P1/19]

A light wave of amplitude  $A$  is incident normally on a surface area  $S$ . The power per unit area reaching the surface is  $P$ . The amplitude of the light wave is increased to  $2A$ . The light is then refocused on to a smaller area  $\frac{1}{4}S$ . What is the power per unit area on this smaller area?

- A  $P$       B  $8P$       C  $12P$       D  $16P$

## 20. [TJC/Prelims 2014/P1/21]

The figure below shows a sound wave travelling to the right in air. Air particles at positions P and Q are at the centre of a compression and a rarefaction respectively.

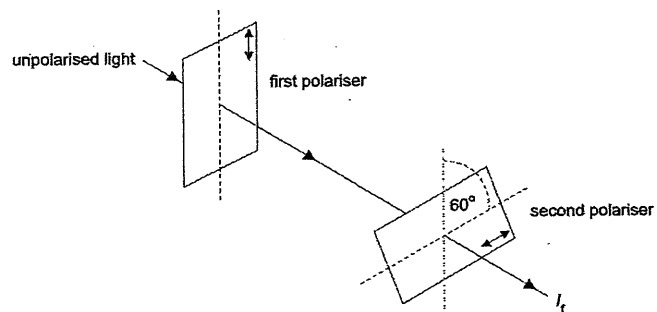


Which row correctly shows the directions of motion of P and Q at the moment shown?

|   | particle P   | particle Q   |
|---|--------------|--------------|
| A | at rest      | at rest      |
| B | at rest      | to the right |
| C | to the left  | to the right |
| D | to the right | to the left  |

## 21. [TJC/Prelims 2014/P1/21]

Unpolarised light is incident on a polariser. The light transmitted by the first polariser is then incident on a second polariser. The polarising axis of the second polariser is at  $60^\circ$  to that of the first polariser.



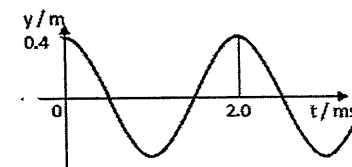
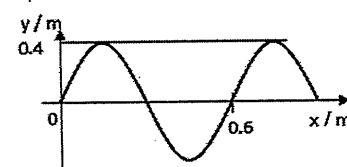
The intensity emerging from the second polariser is  $I_f$ .

Which of the following correctly gives the intensity incident on the first polariser?

- A  $\frac{I_f}{8}$       B  $\frac{I_f}{4}$       C  $4 I_f$       D  $8 I_f$

## 22. [TPJC/Prelims 2014/P1/20]

The displacement-position ( $y$ - $x$ ) graph of a travelling wave at an instant of time and the displacement-time ( $y$ - $t$ ) graph of a particle on the wave are as shown below.



What is the speed of the wave?

- A  $0.2 \text{ m s}^{-1}$       B  $0.3 \text{ m s}^{-1}$       C  $200 \text{ m s}^{-1}$       D  $300 \text{ m s}^{-1}$

## 23. [TPJC/Prelims 2014/P1/21]

A point source emits  $50.0 \text{ W}$  of sound. A small microphone of area  $0.85 \text{ cm}^2$  detects the sound at  $4.0 \text{ m}$  from the source.

What is the power detected by the microphone?

- A  $1.6 \times 10^{-5} \text{ W}$       B  $2.1 \times 10^{-5} \text{ W}$       C  $2.1 \times 10^{-1} \text{ W}$       D  $2.5 \times 10^{-1} \text{ W}$

## 24. [VJC/Prelims 2014/P1/19]

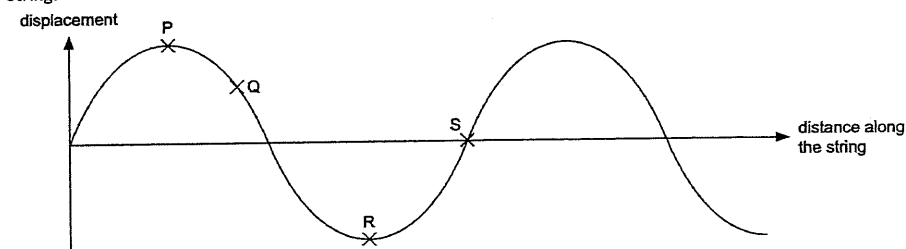
A longitudinal wave of frequency  $1.6 \text{ Hz}$  travels along a stretched spring at a speed of  $2.4 \text{ m s}^{-1}$ .

What is the phase difference between points on the spring that are  $0.50 \text{ m}$  apart?

- A  $\frac{\pi}{3} \text{ rad}$       B  $\frac{2\pi}{3} \text{ rad}$       C  $\pi \text{ rad}$       D  $3\pi \text{ rad}$

## 25. [VJC/Prelims 2014/P1/20]

The figure shows the shape at a particular instant of part of a transverse wave travelling from left to right along a string.

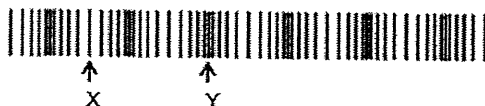


Which statement about the motion of elements of the string at this instant is correct?

- A The speed of Q is higher than S.  
 B Both Q and R are moving upwards.  
 C The energy of P and S is entirely kinetic.  
 D The acceleration of P and R is a maximum.

## 26. [YJC/Prelims 2014/P1/19]

The diagram below represents a 200 kHz longitudinal wave travelling from the left to the right. Two particles in the medium are labelled X and Y.



If the distance between X and Y is 1 mm, what is the speed of the wave in the medium?

- A  $133 \text{ m s}^{-1}$       B  $200 \text{ m s}^{-1}$       C  $267 \text{ m s}^{-1}$       D  $300 \text{ m s}^{-1}$

## 27. [YJC/Prelims 2014/P1/20]

A point source emits 60.0 W of sound in all directions. A small microphone of area  $0.75 \text{ cm}^2$  detects the sound at 5.0 m from the source.

What is the power detected by the microphone?

- A  $1.4 \times 10^{-5} \text{ W}$       B  $1.4 \times 10^{-4} \text{ W}$       C  $9.0 \times 10^{-4} \text{ W}$       D  $1.4 \times 10^{-1} \text{ W}$

## II. Structured Questions

## 1. [CJC/Prelims 2014/P2/2 Part]\*

- a. Explain what is meant by a progressive wave. [1]
- b. Sound is propagated in air as a longitudinal progressive wave, in which there is a repeated sequence of displacements of the air particles. Fig. 1.1a illustrates nine particles, equally spaced along the line AB, in still air.

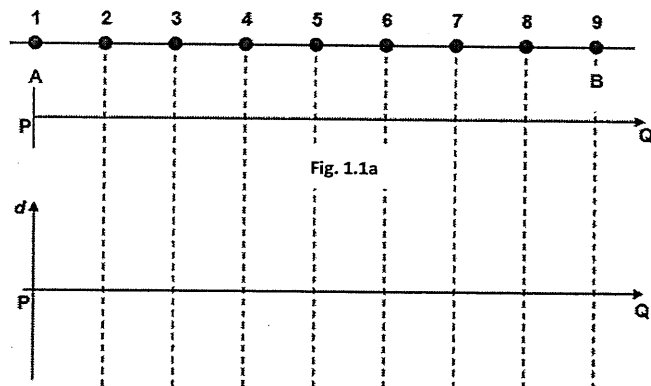


Fig. 1.1b

- i. A sound wave of wavelength equal to the distance between A and B is sent through the air in the direction of PQ.

On line PQ on Fig. 1.1a, draw the possible positions of the nine particles in the wave relative to their undisturbed positions which they occupy in still air, when the sound wave propagates through the particles.

[1]

- ii. Using b. i., sketch a graph showing how the displacement  $d$  of the particles from their undisturbed positions vary along PQ on Fig. 1.1b. Take direction to the right as positive [1]
- iii. A sound wave can also be described in terms of a repeated sequence of changes in pressure. On Fig. 1.1b, identify, and label with H, a point where the pressure is the highest. Justify your answer. [2]
- iv. A loudspeaker, generating a sound wave with a wavelength magnitude equal to AB, and a screen are placed at A and B respectively. Describe and explain the changes, if any, to Fig. 1.1b over time as compared to the current scenario. [2]

[Total: 7]

## 2. [DHS/Prelims 2014/P3/2]\*

The variation with time  $t$  of the displacement  $x$  of a point in a transverse wave  $T_1$  is shown in Fig. 2.1.

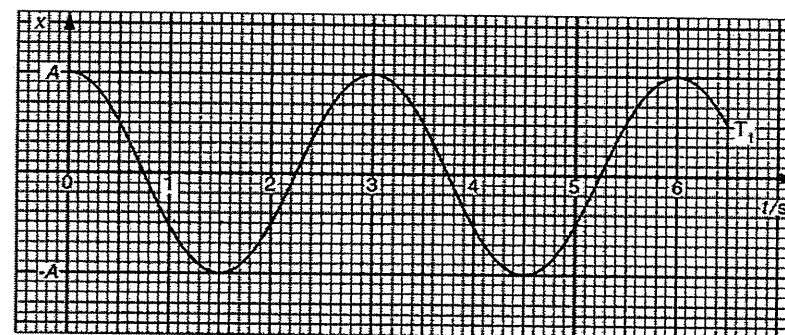


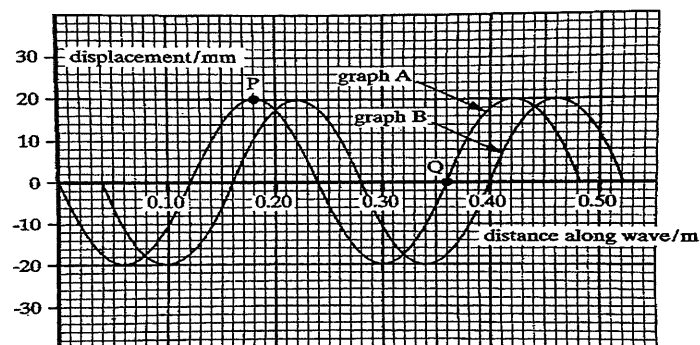
Fig. 2.1

- a. By reference to displacement and direction of travel of wave energy, explain what is meant by a transverse wave. [1]
- b. A second transverse wave  $T_2$  of amplitude  $A$  has the same waveform as wave  $T_1$  but leads  $T_1$  by a phase angle of  $\frac{\pi}{3}$ . The two waves  $T_1$  and  $T_2$  pass through the same point.
- i. On Fig. 2.1, draw the variation with time  $t$  of the displacement  $x$  of the point in wave  $T_2$ . [2]
- ii. Explain what is meant by the principle of superposition of the two waves. [2]
- iii. For the time  $t = 1.0 \text{ s}$ , use Fig. 2.1 to determine, in terms of  $A$
1. the displacement due to wave  $T_1$  alone [1]
  2. the displacement due to wave  $T_2$  alone [1]
  3. the resultant displacement due to both waves. [1]

[Total: 8]

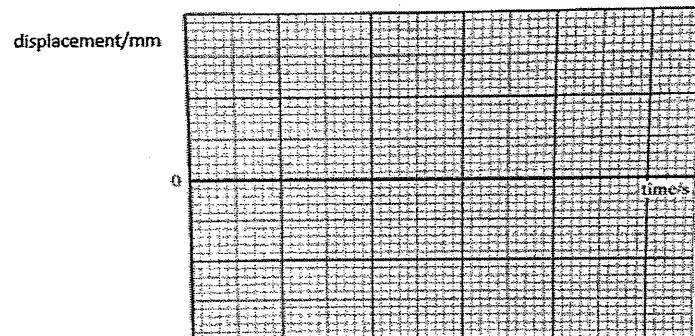
## 3. [NIC/Prelims 2014/P3/7 Part]

- a. Graph A shows the displacement at time  $t = 0$  s at different points on a progressive wave. Graph B shows the situation at  $t = 0.10$  s.



Use graphs A and B to determine, for the wave

- the wave speed [2]
- the phase difference between points P and Q marked on graph A [1]
- Using the grid below, sketch a graph to show the variation with time of the displacement of point P, showing numerical scales on both axes. [3]



- When a stone is thrown into the pond, circular ripples spread out. How could you tell from observations of the ripples, whether or not they were travelling at constant speed? Explain your answer. [2]
- Describe an experiment to demonstrate that light waves are transverse. [3]

[Total: 11]

## 4. [RVHS/Prelims 2014/P3/8 Part]\*

Fig. 4.1 shows the variation with time  $t$  of the displacement  $D_A$  and  $D_B$  at a point X of two sound waves A and B.

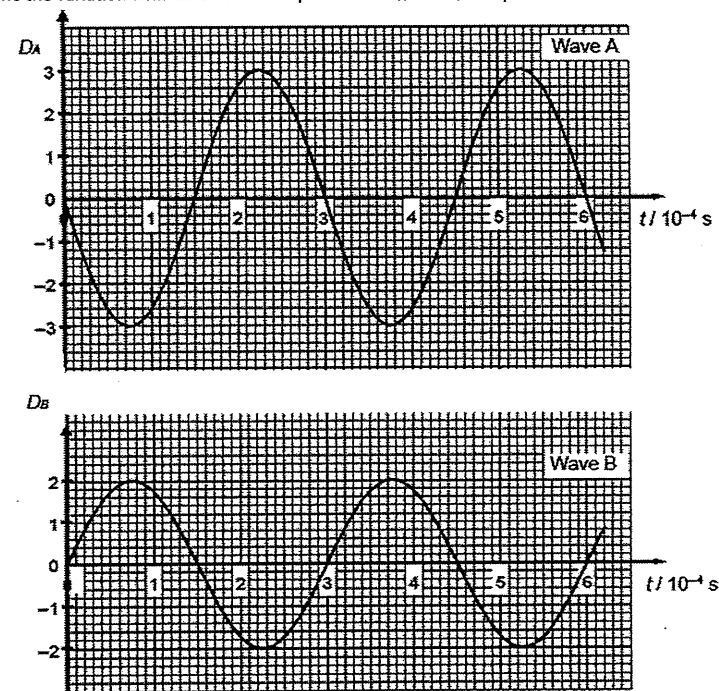


Fig. 4.1

- By reference to Fig. 4.1, state one similarity and one difference between these two waves. [2]
- State, with a reason, whether the two waves are coherent. [1]
- The intensity of wave A alone at point X is  $I$ .
  - Determine the intensity of wave B alone at point X. [2]
  - Determine the resultant intensity of the two waves at point X. [2]

- d. A glass tube, closed at one end, has fine dust sprinkled along its length. A sound source of wave B is placed near the open end of the tube, as shown in Fig. 4.2.

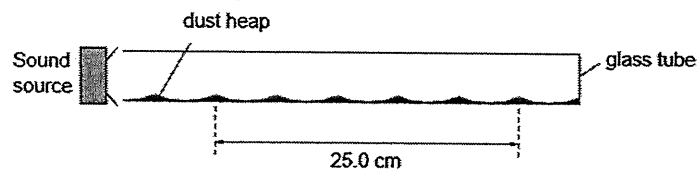


Fig. 4.2

Heaps of dust form in the glass tube.

- Explain, by reference to the properties of stationary waves, why the heaps of dust are formed. [2]
- The distance between six heaps, as shown in Fig. 4.2, is 25.0 cm. Calculate the speed of sound in the tube. [3]
- The wave in the tube is a stationary wave. Explain with reference to the formation of a stationary wave, what is meant by the speed calculated in d. ii.. [2]

[Total: 14]

5. [TPJC/Prelims 2014/P2/3 Part]

- a. A progressive sound wave of wavelength  $\lambda$  and frequency  $f$  passes through a solid from left to right. In Fig. 5.1, the dots on line X represent the equilibrium positions of the atoms in solid and the dots on line Y represent the positions of the same atoms at time  $t_0$ .

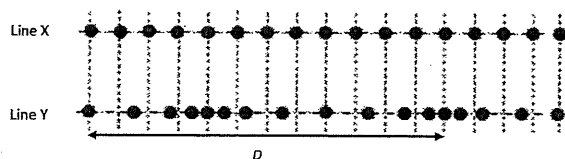


Fig. 5.1

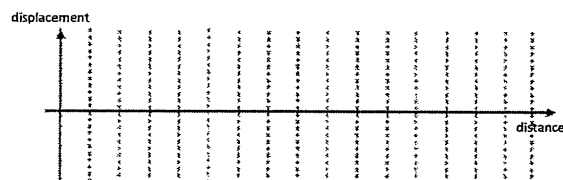


Fig. 5.2

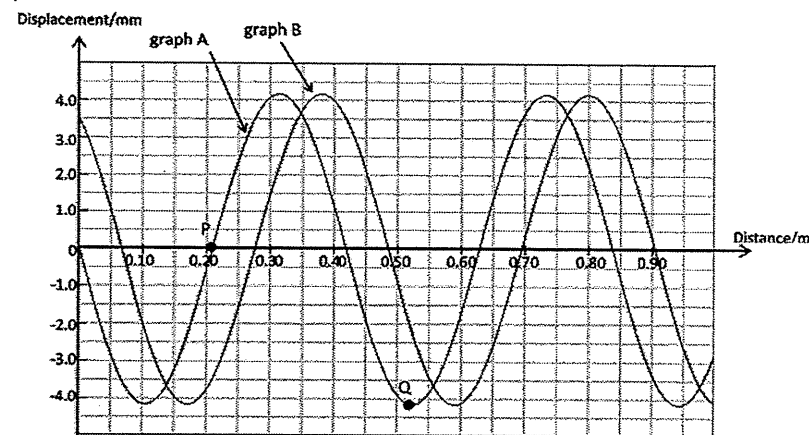
- State what distance  $D$  in Fig. 5.1 represents in terms of  $\lambda$ . [2]
- Taking displacement to the right of equilibrium position as positive, sketch on Fig. 5.2, a graph of displacement against distance at time  $t_0$  for Line Y. [1]

[Total: 3]

6. [VJC/Prelims 2014/P2/5 Part]

- a. Graph A shows the displacement at time  $t = 0$  at different points on a longitudinal progressive sound wave that is travelling to the right. Positive displacement is taken to be displaced to the right.

Graph B shows the wave at  $t = 0.20$  ms.



- Explain what is meant by longitudinal progressive wave. [2]
- Mark on graph B
  - a point C where it is a region of compression (high pressure) [1]
  - a point N where it is a region of normal pressure. [1]
- Use graphs A and B to determine, for the wave
  - the wavelength [1]
  - the wave speed [2]
  - the frequency of the wave [1]
  - the phase difference between points P and Q marked on graph A. [2]

[Total: 10]



### III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

#### MCQs

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| C | D | D | A | A | C | D | A | C | A  | A  | A  | C  | B  | D  |

|    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|
| 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| D  | D  | B  | D  | D  | D  | D  | B  | B  | D  | A  | A  |

- 1 C  $1\lambda = 2\text{ cm}$ , hence  $2.5\text{ cm}$  is  $1.25\lambda$   
phase difference  $= 0.25 \times 2\pi = 0.5\pi$
- 2 D  $I \propto A^2 f^2$   
Amplitude of X  $= 2 \times$  Amplitude of Y  
Period of X  $= 3 \times$  Period of Y  $\Rightarrow$  frequency of X  $= 1/3$  frequency of Y  
 $I_X / I_Y = (2)^2 (1/3)^2$   
 $I_Y = 9/4 I_X = 2.25 I_0$
- 3 D At the original position, the light from Q is polarized vertically. Hence no light will pass through R as the incoming light from Q is in the plane that is perpendicular to the axis of polarisation of R.  
At the final position where the axes of polarisation of Q and R are aligned, the light that passes through P is polarized vertically. Hence no light will pass through Q as the incoming light from P is in the plane that is perpendicular to the axis of polarisation of Q. Therefore the intensity of light through R is zero.  
When Q is rotated, the light intensity will increase until it reaches a maximum at an angle of  $45^\circ$ , and it reduces again as the angle is increased.
- 4 A  $I = I_0 \cos^2 \theta = I_0 \cos^2 50^\circ = 0.413 I_0$
- 5 A Option A is the answer as when  $\theta$  is  $0^\circ$ , the transmission axes are perpendicular to one another and hence no light passes through.  
When  $\theta$  is  $90^\circ$ , the axes are parallel to one another and maximum intensity of light passes through.  
In these two angles, the intensity transmitted increases.

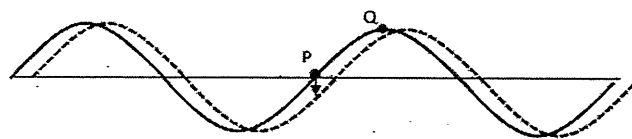
#### Raffles Institution

#### Revision Package (H2 Physics)

#### 10 – Wave Motion

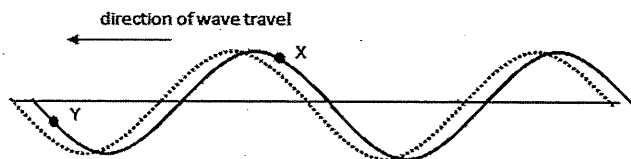
- 6 C By definition, a progressive wave is a wave that transfers energy from one region to another.
- 7 D  $\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda}$   
 $\frac{\Delta\phi}{2\pi} = \frac{3/4\lambda}{\lambda}$   
 $\Delta\phi = \frac{3}{2}\pi \text{ rad}$   
 $\Delta\phi = 270^\circ$
- 8 A  $I \propto A^2$   
 $\frac{I}{I'} = \left(\frac{A}{A'}\right)^2$   
 $\frac{I}{I'} = \left(\frac{A}{A \cos 60^\circ}\right)^2$   
 $\frac{I}{I'} = \left(\frac{1}{0.5}\right)^2$   
 $I' = 0.25 I$
- 9 C Energy  $\propto$  (amplitude) $^2$ , increase amplitude would increase the energy  
Keeping frequency constant so that resonance with water molecules is maintained.
- 10 A In 2 s, the wave travels 1 cm  $\Rightarrow$  speed  $= 1\text{ cm} / 2\text{ s} = 0.5\text{ cm s}^{-1}$
- 11 A For Alice to hear the music as loud as before, the intensity of the sound reaching her ears should remain the same.  
 $\frac{P}{4\pi(15)^2} = \frac{0.75P}{4\pi(x)^2} \Rightarrow x = 13\text{ m}$
- 12 A phase difference  $= \frac{0.22 \times \sin 25^\circ}{1.7} \times 2\pi = 0.34\text{ rad}$
- 13 C  $\lambda = 0.40\text{ m}$   
 $v_{\min} = \frac{0.10}{0.025} = 4.00\text{ m s}^{-1}$   
 $f_{\min} = \frac{4.00}{0.40} = 10\text{ Hz}$   
 $T_{\max} = \frac{1}{10} = 0.10\text{ s}$

14 B



Q is at the maximum displacement and thus is instantaneously at rest.

15 D



16 D

Intensity  $\propto A^2$  and Intensity  $\propto \frac{1}{r}$  for ripples on water

For the same amplitude when power of dipper is doubled,

$$\frac{k}{r_1} = \frac{2k}{r_2}$$

$$r_2 = 2r_1 = 160 \text{ mm}$$

17 D

$$f = \frac{1}{T} \Rightarrow T = \frac{1}{12.5} = 0.08 \text{ s}$$

Consider particle Q in a simple harmonic motion about its equilibrium position with a period of 0.08 s. Q will take 0.02 s to return to its equilibrium position, and another 0.02 s to reach its maximum negative displacement and 0.02 s to return to equilibrium position. Thus option D is correct.

18 B

Since  $I = I_0 \cos^2 \theta$ , graph B is the only cosine square graph.

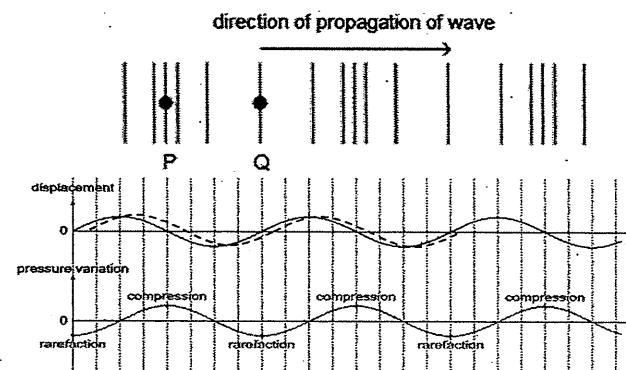
19 D

When light wave has amplitude  $A$ , energy reaching surface area is proportional to  $A^2$  and hence  $P$ , power per unit area is proportional to  $A^2/S$ .

When light wave has amplitude  $2A$ , energy reaching surface area is proportional to  $4A^2$  and power per unit area is proportional to  $4A^2/(1/4S) = 16 A^2/S$ .

20 D

Draw the wave profile at this instant and a short while later to determine the instantaneous velocity.



21 D

|           | unpolarized light            | after 1 <sup>st</sup> polarizer | after 2 <sup>nd</sup> polarizer |
|-----------|------------------------------|---------------------------------|---------------------------------|
| amplitude | Not applicable               | $A$                             | $A_2 = A \cos 60^\circ = 0.5 A$ |
| intensity | $I_0 = 2 \times I_1 = 8 I_f$ | $I_1 = k \times A^2 = 4 I_f$    | $I_f = k \times 0.25 A^2$       |

22 D

$$T = 2.0 \times 10^{-3} \text{ s and } \lambda = 0.6 \text{ m}$$

$$v = f\lambda = \frac{\lambda}{T} = 300 \text{ m s}^{-1}$$

23 B

Since Intensity is the power per unit area, multiply the fraction of area of the microphone to the area of the sphere by the total power of the source.

$$P = \frac{\left(\frac{0.85}{100^2}\right)}{4\pi(4.0)^2} \times 50.0 = 2.11 \times 10^{-5} \text{ W}$$

24 B

$$v = f\lambda \Rightarrow 2.4 = 1.6\lambda \Rightarrow \lambda = 1.5 \text{ m}$$

$$\phi = \frac{x}{\lambda} \times 2\pi = \frac{0.50}{1.5} \times 2\pi = \frac{2\pi}{3} \text{ rad}$$

25 D

Option D is correct. Since all particles on the string are vibrating in SHM, at P and R the particles are at the maximum displacements and thus will have maximum accelerations.

Option A is wrong because S is at the equilibrium position and has maximum speed and thus higher speed than Q.

Option B is wrong because R is momentary at rest and hence not moving.

Option C is wrong because P has entirely potential energy, not KE.

- 26 A distance between 2 adjacent rarefactions or 2 adjacent compressions is the wavelength.

$$1.5\lambda = 1 \text{ mm} \Rightarrow \lambda = 0.667 \text{ mm}$$

$$v = f\lambda = 200000 \times \frac{0.667}{1000} = 133 \text{ m s}^{-1}$$

- 27 A

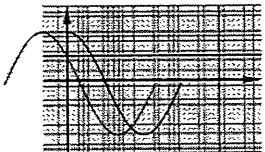
$$P = \frac{\left(\frac{0.75}{100^2}\right)}{4\pi(5.0)^2} \times 60.0 = 1.4 \times 10^{-5} \text{ W}$$

### Structured Questions

1. [CJC/Prelims 2014/P2/2 Part]\*

- A progressive wave transports energy from one point to another in the direction of wave propagation.
- For part i., positions of particles must conform to one wavelength, with clear differences in displacements of particles.
  - For part ii., wave drawn must correspond to diagram in i. and displacement of particles to the right is taken to be positive on the graph.
  - Point H, a point of high pressure, occurs at where particle 5 is as this is where compressions occur due to neighbouring air particles 4 and 6 coming closer to one another.
  - As a stationary wave is being formed, the wave will not progress and each particle will oscillate with different amplitudes.  
In the current scenario, the wave will progress towards Q with each particle moving left and right, reaching the same amplitude over time.

2. [DHS/Prelims 2014/P3/2]\*

- displacement and direction of energy travel are perpendicular to one another
- 

- See definitions in lecture notes

- $-\frac{1}{2}A$
  - $-A$
  - $-\frac{3}{2}A$

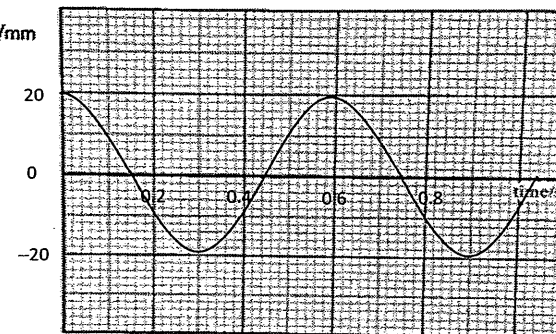
3. [NJC/Prelims 2014/P3/7 Part]

- From the graph, distance moved by wave = 0.040 m  
Time taken = 0.10 s  
Speed of wave = distance / time = 0.040 / 0.10 = 0.40 m s<sup>-1</sup>

- Phase difference =  $2\pi \times \text{path difference} / \text{wavelength} = 2\pi \times 0.18 / 0.24 = 3\pi / 2 \text{ rad}$

- $v = f\lambda \Rightarrow 0.40 = \left(\frac{1}{T}\right) \times 0.24 \Rightarrow T = 0.60 \text{ s}$

displacement/mm



- Constant wavelength ( $\lambda$ ) must be observed at different points along the wave propagation. Since the frequency of the ripple is constant, the speed of the wave given by  $v = f\lambda$  will be constant.
- Transverse wave can be polarized. Longitudinal wave cannot be polarized.  
Pass a beam of light through 2 consecutive polarisers with their transmission axis parallel to one another. Then slowly rotate the axis of the second polariser to be at 90° to the axis of the first polariser. By observing the other end of the pair of polarisers, it can be observed that the intensity of the beam of light goes from bright to zero. This shows that light waves are transverse.

4. [RVHS/Prelims 2014/P3/8 Part]\*

- Same frequency / period / type  
Different amplitude / phase
- Yes. They have constant phase difference and are in anti-phase (phase difference of  $\pi$  rad).

$$\begin{aligned} \text{c. i. } I &\propto A^2 \\ \frac{I_B}{I_A} &= \frac{A_B^2}{A_A^2} = \frac{2^2}{3^2} \\ I_B &= \frac{4}{9} I = 0.44 I \end{aligned}$$

- $I \propto A^2$   
Since the two waves are in anti-phase, they will undergo destructive interference,  
Resultant amplitude =  $3 - 2 = 1$  unit.

$$\begin{aligned} \frac{I_R}{I_A} &= \frac{A_R^2}{A_A^2} = \frac{1^2}{3^2} \\ I_R &= \frac{1}{9} I = 0.11 I \end{aligned}$$

- Stationary sound wave is a longitudinal wave where particles oscillate parallel to the length of the tube.  
The antinodes (maximum displacement amplitude) push the dust to region of nodes (minimum displacement amplitude) of the stationary wave. Thus heaps of dust formed at region of the nodes.

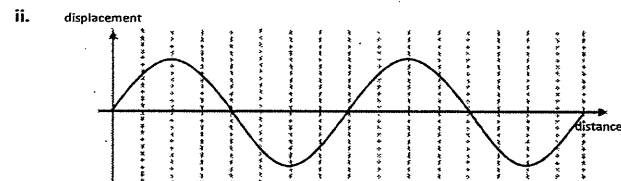
ii. 
$$v = f\lambda = \frac{1}{3 \times 10^{-4}} \times \frac{0.25}{2.5} = 330 \text{ m s}^{-1}$$

- iii. When the progressive wave travelling down the tube and its reflection meet and undergo superposition with each other, a stationary wave is formed.

The speed is the speed of the progressive sound wave in the tube.

5. [TPJC/Prelims 2014/P2/3 Part]

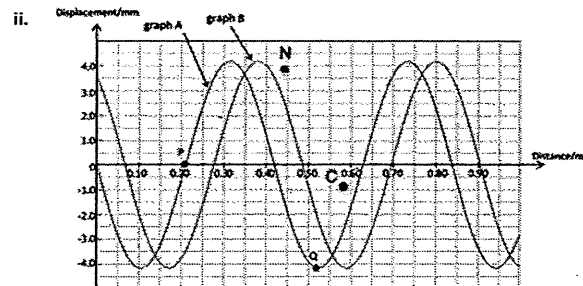
- a. i. 1.5 wavelengths (or  $1.5\lambda$ ).



6. [VJC/Prelims 2014/P2/5 Part]

- a. i. A longitudinal wave is one in which its particles oscillate in a direction parallel to the direction of energy transfer.

A progressive wave transports energy from one point to another in the direction of wave propagation.



- iii. 1. From the graph, wavelength  $\lambda = 0.42 \text{ m}$ .
2. From graph A to graph B, the wave has moved a distance of  $0.070 \text{ m}$  in  $0.20 \text{ ms}$ .

$$v = \text{distance} / \text{time} = 0.070 / (0.20 \times 10^{-3}) = 350 \text{ m s}^{-1}$$

3.  $v = f\lambda$

$$f = v/\lambda = 350/0.42 = 830 \text{ Hz}$$

4. distance between P and Q,  $x = 0.31 \text{ m}$

$$\text{phase diff, } \phi = (x/\lambda) \times 2\pi = (0.31/0.42) \times 2\pi = 4.6 \text{ rad.}$$

Or

P and Q is  $\frac{3}{4}$  of a wavelength apart, hence

$$\phi = (x/\lambda) \times 2\pi = (\frac{3}{4}) \times 2\pi = 3\pi/2$$